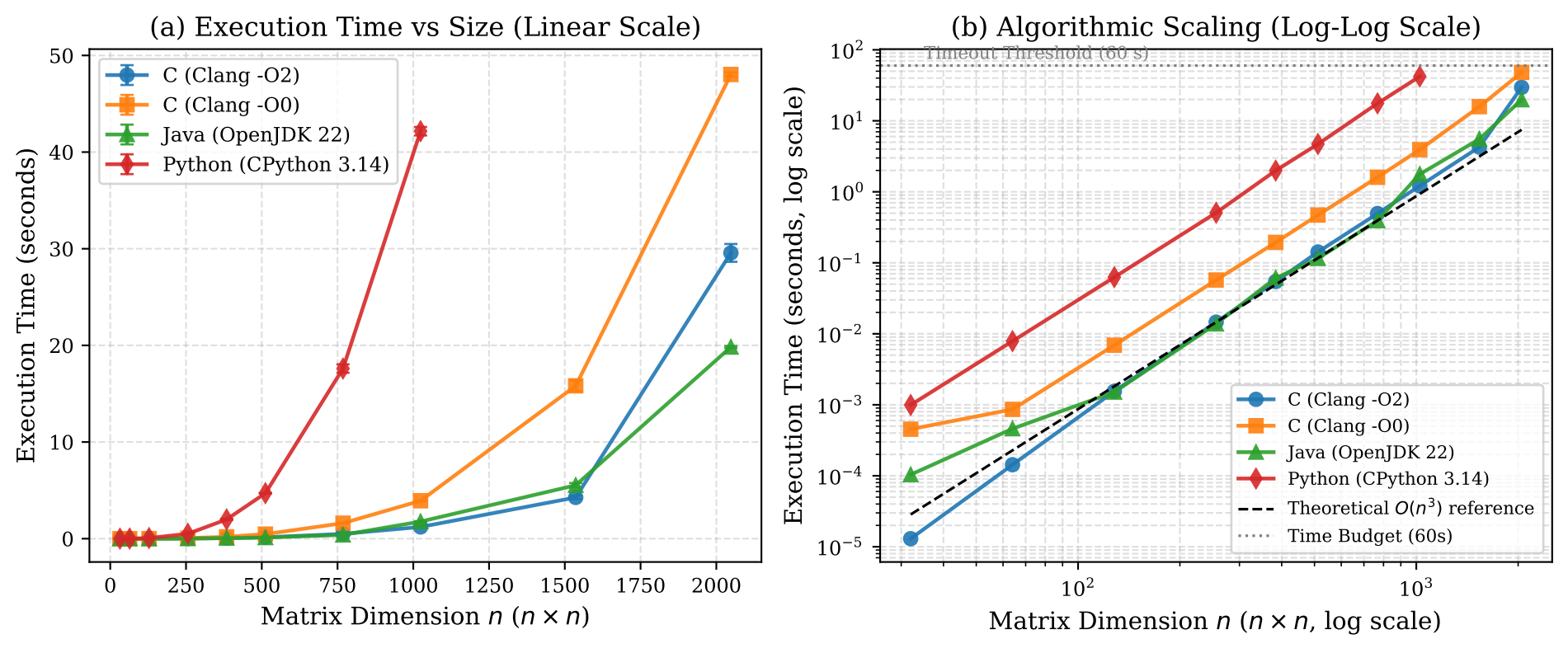
(a) Execution Time vs Size (Linear Scale)
Execution Time (seconds)
Matrix Dimension n ($n \times n$)
C (Clang -O2)
C (Clang -O0)
Java (OpenJDK 22)
Python (CPython 3.14)
0
10
20
30
40
50
0
250
500
750
1000
1250
1500
1750
2000
(b) Algorithmic Scaling (Log-Log Scale)
Timeout Threshold (60 s)
Execution Time (seconds, log scale)
Matrix Dimension n ($n \times n$, log scale)
C (Clang -O2)
C (Clang -O0)
Java (OpenJDK 22)
Python (CPython 3.14)
Theoretical $O(n^3)$ reference
Time Budget (60s)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{2}
10^{3}